

SESSION 2

RAD-423 Imaging Informatics

PACS, RIS, Storage Systems & Information Management

How medical images and radiology information are stored, retrieved, protected and coordinated across the workflow

Course	RAD-423 Imaging Informatics
Date	Sept, 2026
Lecturer	A. A. Zubair
Suggested study time	60–90 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. explain why medical imaging requires organized storage, retrieval and information-management systems;
2. describe the main functions of PACS and distinguish PACS from RIS, HIS and EHR;
3. compare direct-attached storage, network-attached storage and storage-area networks;
4. explain the purpose of a Vendor Neutral Archive and storage tiering;
5. describe the 3-2-1 backup concept and distinguish backup from routine storage redundancy; and
6. explain the radiographer's role in maintaining accurate, retrievable and correctly linked imaging information.

Key Concept

An imaging department needs more than scanners and image files. It needs a coordinated information system that can identify the patient and examination, receive images, store them reliably, retrieve them quickly, display them to authorized users and preserve them for future comparison.

1 Why Imaging Data Management Matters

Modern radiology produces large volumes of digital information. A CT examination may contain hundreds or thousands of images; an MRI examination may contain multiple sequences; ultrasound can generate still images, cine loops and measurements; nuclear medicine can generate dynamic, planar and tomographic datasets. Once

acquired, these data must remain associated with the correct patient and examination and must be available when needed.

For the imaging department, information management can be understood through four practical questions:

1. **Where will the study be stored?**
2. **How will an authorized user find it again?**
3. **How will the image data remain linked to the correct clinical and examination information?**
4. **What happens if a server, disk, network or site fails?**

Clinical / Practical Link

A previous CT chest may be clinically important years later because the radiologist may need to compare a new pulmonary nodule with an older study. Long-term storage is therefore not simply an IT convenience; it supports continuity of care and comparison imaging.

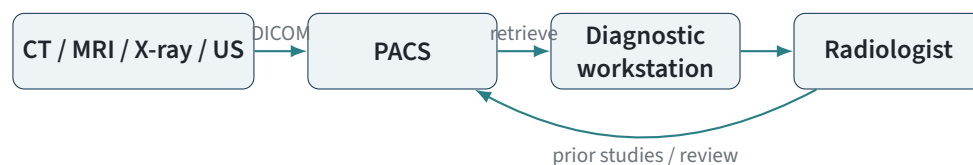
2 PACS: Picture Archiving and Communication System

PACS stands for **P**icture **A**rchiving and **C**ommunication **S**ystem. Its core role is the management of medical images and associated imaging information. PACS enables images to be received from modalities, stored, retrieved, distributed and displayed.

Six central PACS functions are:

- image storage;
- image retrieval;
- image distribution;
- image viewing;
- image management; and
- communication between imaging devices and workstations.

2.1 A simplified PACS workflow



The scanner does not need to keep the only copy of the examination. After transmission, PACS becomes a central point from which authorized workstations can retrieve the study. This reduces dependence on physical film and allows images to be viewed in multiple locations.

High-Yield

PACS is not simply a large hard drive. A complete PACS combines **storage, indexing, retrieval, distribution, display and workflow support**. The ability to locate the correct study is as important as the ability to store its pixel data.

2.2 Typical components of a PACS environment

A practical PACS environment commonly includes:

- imaging modalities that send studies;
- PACS application servers or services;
- image storage or archive systems;
- a database or index describing stored studies;
- diagnostic and clinical viewing workstations;
- the network infrastructure connecting these components; and
- interfaces to radiology and hospital information systems.

If the images are present on storage but the database cannot correctly identify or retrieve them, the system is not functioning adequately from the user's perspective. PACS therefore depends on both **data storage** and **information management**.

3 RIS, HIS and EHR

PACS mainly manages imaging objects. Other systems manage the workflow and broader patient record.

3.1 Radiology Information System (RIS)

A **Radiology Information System (RIS)** is primarily concerned with radiology workflow and examination information. Typical RIS functions include:

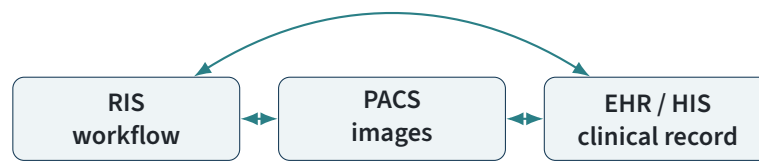
- patient registration;
- examination scheduling;
- appointment management;
- worklists;
- examination tracking;
- reporting;
- result distribution; and
- billing-related or administrative information, where applicable.

The RIS helps answer questions such as: *Who is the patient? What examination was requested? When is it scheduled? Has the examination been performed? Has the report been completed?*

3.2 Hospital Information System and Electronic Health Record

A **Hospital Information System (HIS)** supports information management across the wider hospital. An **Electronic Health Record (EHR)** contains broader longitudinal clinical information about the patient, which may include demographics, diagnoses, laboratory results, medications, radiology reports and links to imaging

information.



In practice, these systems work together. A radiologist may retrieve images from PACS while simultaneously viewing the request, clinical indication and previous report information supplied through integrated RIS/EHR systems.

3.3 PACS versus RIS

Feature	PACS	RIS
Primary emphasis	Medical-image management	Radiology workflow and examination information
Typical content	Images and associated imaging objects	Scheduling, status, worklists, reports, administrative data
Common user task	Retrieve and display a study	Track and manage an examination
Relationship to DICOM	Strongly image/DICOM oriented	Exchanges workflow information and interfaces with imaging systems
Clinical role	Makes imaging studies available for interpretation and review	Coordinates the radiology process from request to completion/reporting

Common Error

Do not write that PACS and RIS are the same system simply because some modern products present them through one interface. Their conceptual roles remain different: **PACS manages images; RIS manages radiology workflow information**. Integration does not erase the distinction.

4 Data Storage Architectures

Medical imaging requires storage that can cope with large files, repeated access and long retention periods. Three important storage architectures are DAS, NAS and SAN.

4.1 Direct Attached Storage (DAS)

Direct Attached Storage is storage connected directly to a computer or server rather than accessed as a shared storage service across a network.

Advantages

- simple to understand and deploy;
- often relatively inexpensive for small systems; and
- does not require a separate storage network.

Limitations

- limited scalability;
- more difficult to share between many systems; and
- the host computer or server may become a single point of dependence.

DAS can be useful in small or specialized environments, but enterprise imaging departments usually require more flexible shared storage.

4.2 Network Attached Storage (NAS)

Network Attached Storage provides storage over a network. Multiple authorized systems can access centrally managed files or storage resources.

Its advantages include easier sharing, centralized management and scalability. Its performance and availability, however, depend on the network and the correct configuration of security and access controls.

4.3 Storage Area Network (SAN)

A **Storage Area Network** is a dedicated high-speed storage environment designed to provide servers with access to shared storage resources. SANs are common in enterprise settings because imaging organizations may generate very large volumes of data and may require predictable high-throughput storage access.

	DAS	NAS	SAN
Connection	Direct to one host/server	Shared over a data network	Dedicated storage network / fabric
Sharing	Limited	Straightforward file sharing	Designed for high-performance shared storage
Scalability	Usually lower	Moderate to high	High in enterprise environments
Typical use	Small/local systems	Departmental/shared storage	Large enterprise workloads
Main dependency	Host device	General network	Storage network and enterprise infrastructure

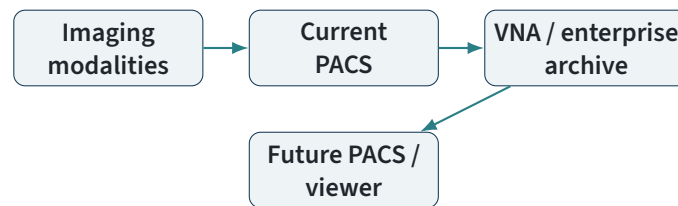
Why imaging storage grows quickly

Imagine a hospital operating several CT scanners, MRI scanners, digital radiography rooms and ultrasound systems. Every examination adds new image data, and many studies must be retained for years. Even when a single examination seems modest, the accumulated daily, monthly and yearly volume can become very large. This is why storage planning must consider future growth rather than today's workload only.

5 Vendor Neutral Archive (VNA)

A **Vendor Neutral Archive (VNA)** is an enterprise archive intended to preserve imaging data independently of a single PACS vendor's proprietary environment. The purpose is not that every system becomes technically identical, but that the organization reduces dependence on one PACS product for long-term access to historical imaging information.

The practical reason for this is straightforward: hospitals may replace or upgrade PACS platforms. If historical data are tightly tied to the old vendor's system, migration can be costly and complicated. A VNA can provide a more stable enterprise archive layer beneath changing viewing or workflow systems.

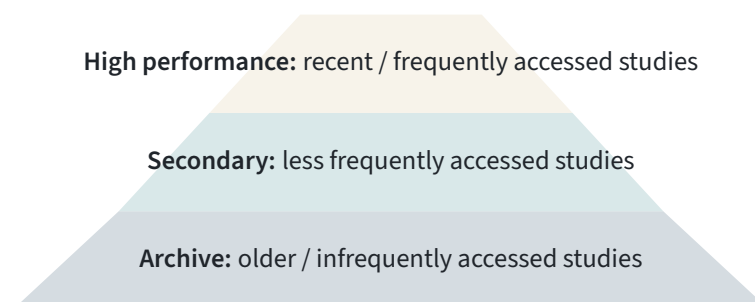


High-Yield

A VNA and a PACS are related but not identical. PACS focuses strongly on image workflow, retrieval and display; a VNA is primarily an enterprise archive strategy intended to preserve imaging information across system and vendor changes.

6 Storage Tiers

Not every study needs the same storage performance forever. Imaging organizations may therefore use **storage tiers**.



A simple three-tier model is:

Tier 1 – high-performance storage

Used for recent or frequently accessed studies where rapid retrieval is important.

Tier 2 – secondary storage

Used for studies that are less frequently requested but still need reasonably convenient access.

Tier 3 – archive

Used for long-term retention of older historical studies.

The boundaries between tiers depend on local technology, policy and workload. The central idea is to balance **performance, capacity and cost**.

7 Backup and Disaster Recovery

Storage and backup solve different problems. Storage keeps the working data. Backup creates additional copies that can be used to recover information after loss or damage.

A useful approach to backup planning is the **3-2-1 backup strategy**:

- keep **3 copies** of important data;
- use **2 different storage media or storage types**; and
- keep **1 copy off-site**.



This strategy is intended to reduce the chance that one failure destroys every usable copy. Threats include hardware failure, accidental deletion, cyberattack, fire, flood or a wider site disaster.

7.1 Redundancy Is Not the Same as Backup

Some storage systems use disk redundancy so that the system can continue operating after a disk failure. This is useful for availability, but it should not automatically be called a backup. If the active data are deleted, corrupted or encrypted by malware, a redundant storage array may faithfully reproduce the same problem across its disks.

Common Error

“We use RAID, therefore we have a backup” is incorrect. Redundancy helps tolerate some hardware failures; backup provides separate recoverable copies. A robust department may need both.

7.2 Recovery Objectives

Two useful disaster-recovery ideas are:

Recovery Point Objective (RPO)

How much recent data loss the organization is prepared to tolerate, usually expressed as a time interval.

Recovery Time Objective (RTO)

How long the service may remain unavailable before it must be restored.

These concepts help translate the vague instruction “have a backup” into operational questions: *How recent must the restored data be, and how quickly must the imaging service return?*

8 The Radiographer’s Role in Information Management

Radiographers interact with these systems every day. The responsibilities are not limited to producing a diagnostic image.

Important responsibilities include:

- confirming the correct patient and examination before acquisition;
- ensuring studies are associated with the correct patient record;
- checking that the examination has been transmitted successfully to PACS;
- recognizing missing, duplicated or incorrectly labelled studies;

- completing examination status correctly according to departmental workflow;
- avoiding unnecessary duplicate acquisition simply because an image is difficult to locate; and
- escalating PACS, RIS or network problems through the correct support pathway.

Clinical / Practical Link

Suppose a radiographer completes an emergency CT but the study fails to reach PACS. The scanner images may be technically excellent, yet the examination cannot support timely reporting until the information-flow problem is recognized and corrected. Informatics competence is therefore part of safe imaging practice.

Key Takeaways

- PACS stores, retrieves, distributes and displays medical images and related imaging information.
- RIS manages radiology workflow, including scheduling, examination tracking and reporting-related information.
- HIS/EHR systems contain broader hospital and patient clinical information and work with PACS/RIS in an integrated environment.
- DAS, NAS and SAN represent different ways of providing storage, with different levels of sharing, scalability and performance.
- A VNA helps preserve enterprise imaging data independently of a single PACS vendor environment.
- Storage tiering balances retrieval performance, capacity and long-term retention needs.
- Backup protects against data loss; routine storage redundancy alone is not equivalent to a backup.
- The 3-2-1 strategy provides a simple framework for maintaining multiple recoverable copies.

Key Terms

PACS

Picture Archiving and Communication System; manages storage, retrieval, distribution and display of medical images.

RIS

Radiology Information System; manages radiology workflow and examination information.

HIS

Hospital Information System; supports information management across the hospital.

EHR

Electronic Health Record; broader longitudinal digital clinical record of the patient.

DAS

Direct Attached Storage; storage connected directly to a host computer or server.

NAS

Network Attached Storage; storage made available over a network.

SAN

Storage Area Network; high-speed shared storage infrastructure used in enterprise environments.

VNA

Vendor Neutral Archive; enterprise imaging archive intended to reduce dependence on one PACS vendor.

Storage tier

A category of storage selected according to performance, access frequency, capacity or retention needs.

Backup

A separate recoverable copy of data maintained to support restoration after loss or damage.

Self-Assessment Questions

1. Define PACS and describe four of its major functions.
2. Distinguish PACS from RIS using two practical examples.
3. Explain how RIS, PACS and EHR can cooperate during a radiology examination.
4. Compare DAS, NAS and SAN in terms of connectivity and scalability.
5. Explain why a large imaging department may use storage tiers rather than one identical storage system for every study.
6. What problem is a Vendor Neutral Archive intended to reduce?
7. State the 3-2-1 backup principle.
8. Explain why storage redundancy should not automatically be treated as backup.
9. Define RPO and RTO and explain why they are useful during disaster-recovery planning.
10. A CT examination was completed but cannot be found on PACS. List the information-management checks a radiographer should make before considering repeat acquisition.

Series Navigation

1	Data Acquisition & DICOM Foundations
2	PACS, RIS, Storage Systems & Information Management — this session
3	Cybersecurity in Medical Imaging
4	Cloud-Based Medical Imaging
5	Health Data Analytics, Interoperability & Applied Practice

